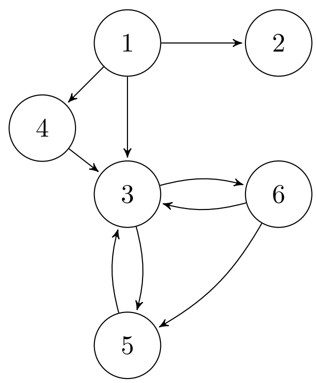
Weekly Assignment 3:
Depth-First Search

September 2024

1. Apply DFS to the graph below using 1 as the start node.
   - Specify the discovered states and the content of the stack at each iteration.
   - Specify for all edges in the graph their type (tree, forward, backward or cross).

   You may assume that when adding nodes to the stack, the vertex with the highest value is inserted *first*.



2. For each of the following statements, explain whether they are true or false.
   (a) If a depth-first search on a directed graph $G = (V, E)$ produces exactly one backward edge, then it is possible to choose an edge $e \in E$ such that the graph $G' = (V, E - \{e\})$ is acyclic.
   (b) If a directed graph G is cyclic but can be made acyclic by removing a single edge, then a depth-first search in G will encounter exactly one backward edge.

3. Suppose a Computer Science curriculum consists of n courses. Each course is mandatory. The prerequisite graph for the curriculum is a directed, acyclic graph (DAG) G that has a node for each course, and an edge from course v to course w if and only if v is a prerequisite for w. In such a case, a student is not allowed to take course w in the same or an earlier semester than she takes course v. A student can take any number of courses in a single semester.

   Design an algorithm to find out the minimum number of semesters necessary to complete the curriculum. The running time of your algorithm should be linear. Discuss the correctness of your solution.

4. Design a linear-time algorithm for the following task:

Input: A directed acyclic graph G

Output: Does G contain a path that visits all vertices of G?

Discuss the time complexity of your algorithm and explain why it is correct.